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CHEMISTRY

CAPS-2$^+$

TARGET : JEE-ADVANCED-2024

MCQ (Multiple Correct Type) :

1. Which of the following reaction(s) can evolve phosphine ?

(A) White P + $Ca(OH)_2 \longrightarrow$ (B) AlP + $H_2O \longrightarrow$

(C) $H_3PO_4 \xrightarrow{\Delta}$ (D) PH_4I + NaOH $\xrightarrow{\Delta}$

2. Which of the following statement(s) is/are correct regarding graphite and inorganic graphite ?

(A) Both are slippery in nature

(B) Both are conductor of electricity but graphite is less conducting in nature compared to inorganic graphite

(C) Graphite is not having any charge separation like inorganic graphite

(D) All atoms in graphite as well as in inorganic graphite are sp^2 hybridised

3. 1 mol red lead (Pb_3O_4) is reacted separately with conc. HCl and conc. HNO_3 solutions under apporpriate conditions. Which of the following option(s) is/are correct ?

(A) A gas is released in reaction with HCl

(B) A gas is released in reaction with HNO_3

(C) Both reactions are redox reactions

(D) The difference in moles of HCl and HNO_3 needed for complete reaction with 1 mol red lead is 4.

4. Select substance(s) which is/are used to convert Mn^{2+} to MnO_4^- ?

(A) H_2O_2/H^+ (B) PbO_2/H^+ (C) $S_2O_8^{2-}$ (D) $Cr_2O_7^{2-}/H^+$

5. What is/are true for hydrogen peroxide and ozone ?

(A) H_2O_2 acts as a stronger reducing agent in alkaline medium than in acidic medium

(B) H_2O_2 and O_3 both are oxidising agents as well as bleaching agents

(C) H_2O_2 has high dielectric constant

(D) Ozone is used in the manufacture of potassium permanganate from pyrolusite

6. Which of the following statement(s) is/are correct ?

(A) $BeCO_3$ is kept in the atomosphere of CO_2 since, it is least thermally stable alkaline earth metal carbonate

(B) Be dissolve in an alkali solution forming $[Be(OH)_4]^{2-}$

(C) BeF_2 form complex ion with NaF in which Be goes with cation

(D) BeF_2 form complex ion with NaF in which Be goes with anion.

7. The possible product(s) formed in the following reaction is/are :

$IF_5 + H_2O \longrightarrow ?$

(A) HIO_3 (B) HIO (C) HIO_4 (D) HF

8. Identify the correct statement(s) regarding caustic soda :

(A) Its solution gives disproportionation reaction with red P

(B) It reacts with CO gas at 150°–200° C, 5–10 atm forming sodium formate

(C) It cannot co-exist with Na_2HPO_3 in aqueous solution

(D) It is deliquescent in nature

9. The boiling point of an aqueous urea solution is 101.04°C. The density of solution is 1.12 gm/ml. Which of the following is/are correct statement(s) regarding this solution ?

[Given : For water : $K_b = 0.52$ K–Kg mol^{-1}, $K_f = 1.86$ K–Kg mol^{-1}**]**

(A) The molarity of solution is 2 M

(B) The freezing point of solution is 3.72°C

(C) If the solution is cooled to –1.86°C, no ice will form

(D) At any temperature, the relative lowering of vapour pressure is $\left(\frac{9}{259}\right)$.

10. Select the incorrect statement(s) :

(A) Liquation process is used for purification of impure tin and zinc metals as the impurities present have lesser melting point than the metal itself.

(B) Fractional crystallization method produces metal of very high purity.

(C) Bessemerisation of cast iron involves oxidative refining.

(D) Cupellation involves removal of Ag impurity from Pb metal.

11. Which of the following reaction(s) is/are used in the preparation of colloidal solution?

(A) $AuCl_3 + SnCl_2 \rightarrow Au + SnCl_4$ (B) $As_2O_3 + H_2S \rightarrow As_2S_3 + H_2O$

(C) $H_2S + SO_2 \rightarrow S + H_2O$ (D) $FeCl_3 + H_2O \rightarrow Fe(OH)_3 + HCl$

12. Which of the following reaction(s) takes place in the Bessemer converter for the extraction of Cu?

(A) $2CuFeS_2 + 4O_2 \rightarrow Cu_2S + 2FeO + 3SO_2$ (B) $CuS + 2O_2 \rightarrow CuO + SO_3$

(C) $Cu_2S + 2Cu_2O \rightarrow 6Cu + SO_2$ (D) $FeO + SiO_2 \rightarrow FeSiO_3$

13. An equimolar mixture of benzene & toluene is prepared. The total vapour pressure of this mixture as fraction of mole of benzene is found to be $P_T = 200 + 400\, X_{benzene}$ mm of Hg. Than :

(A) Pure state vapour pressure of benzene is 600 mm of Hg.

(B) Pure state vapour pressure of toluene is 200 mm of Hg.

(C) Ratio of mole fraction of benzene and toluene in vapour phase after 5 times fractional distillation is $2^5 : 1$.

(D*) Ratio of mole fraction of benzene and toluene in vapour phase after 5 times fractional distillation is $3^5 : 1$.

(Paragraph) :

Answer Q.14, Q.15 and Q.16 by appropriately matching the information given in the three columns of the following table

Column – I (Archimedean solid)		**Column – II**		**Column - III**	
(a)	Truncated octahedron	(P)	14 regular face	(I)	36 edges
(b)	Truncated tetrahedron	(Q)	8 hexagonal face	(II)	18 edges
(c)	Truncated cube	(R)	4 hexagonal face	(III)	12 edges

14. Which of the following is correct combination ?

(A) (a, Q, I) (B) (b, Q, III) (C) (c, R, II) (D) (a, P, III)

15. Which of the following is correct combination ?

(A) (a, R, I) (B) (b, R, II) (C) (c, P, II) (D) (b, Q, III)

16. Which of the following is correct combination ?

(A) (a, P, II) (B) (b, P, III) (C) (c, P, I) (D) (c, R, III)

(Numerical Value type) :

17. When the mixture of two immiscible liquids [water & nitrobenzene) boils at 372 K and the vapour pressure at this temperature are 98.4 $KP_a(H_2O)$ and 3.6 $KP_a(C_6H_5NO_2)$. If the weight ratio of nitrobenzene and water in the vapour is 1 : X, then the value of 'X^2' is.

18. How many of the following statements are correct ?

(1) At critical conditions volume occupied by the gas is 12 times the volume of 1 mole gaseous molecules

(2) A gas can be liquefied at temperature above its T_b (Boyle's temperature) by application of pressure

(3) At very high pressure and moderate temperature the volume of n mole real gas is equal to $\frac{nRT}{P}+nb$

(4) For a real gas following equation $\left(P+\frac{\alpha}{TV_m^2}\right)(V_m - \beta) = RT$ where α & β are constant. Boyle's temperature is $\sqrt{\frac{\alpha}{R\beta}}$

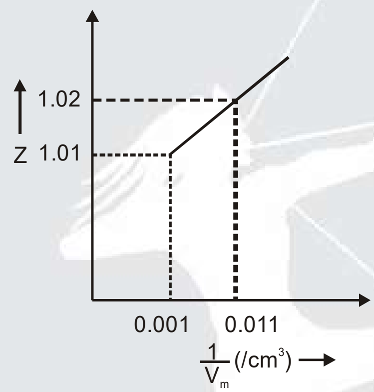
(5) Partial pressure of O_2 gas will increase if He gas is introduced isothermally in a rigid vessel having O_2 and N_2 gases (consider ideal behaviour)

(6) $\frac{P}{V}$ v/s P graph will be a rectangular hyperbola for a fixed amount of an ideal gas at constant temperature

(7) $\frac{V}{P}$ v/s $\frac{1}{P^2}$ graph will be a straight line for a fixed amount of an ideal gas at constant temperature

19. Consider the following graph of Z v/s $\frac{1}{V_m}$ drawn at low pressure and inversion temperature ($T = \frac{2a}{Rb}$), using the virial form of van der Waal's equation :

$$\mathbf{Z = 1 +} \left(b - \frac{a}{RT}\right)\frac{1}{V_m} + \frac{b^2}{V_m^2} + \frac{b^3}{V_m^3} + ...$$



Find critical molar volume (V_C) in cm^3?

20. 10% sites of a catalyst bed have adsorbed H_2. On heating H_2 gas is evolved from sites and collected at 0.03 atm and 300 K in a small vessel of 2.46 cm^3. Number of sites available is 5.4×10^{16} per cm^2 and surface area is 1000 cm^2. Find out the number of surface sites occupied per molecule of H_2 (R = 0.082 atm L/ mol K, $N_A = 6.02 \times 10^{23}$)

21. A solution of crab hemocyanin, a pigmented protein extracted from crabs, was prepared by dissolving 0.750 g in 125 cm^3 of an aqueous medium. At 39°C an osmotic pressure rise of 2.6 mm of the solution was observed. The solution had a density of 1.00 g/cm^3. Determine the molar mass (g/mol) of the protein.

Given : g = 10 m/s^2, R = 25/3 J/mol K

Report your answer after multiplying by 10^{-4}.

22. At room temperature 300 K orange juice gets spoilt in about 51 hours. In a refrigerator at 200 K juice can be stored three times as long before it gets spoilt. Estimate how much time (in hours) should it take for juice to get spoilt at 400 K?

Given : ln 3 = 1.1, $e^{(11/20)} = 1.7$

23. Find total number of metal cations which are precipitated as metal sulphide on passing H_2S gas through metal salt solution.

Pb^{2+}(aq),	Mn^{2+}(aq),	Sn^{2+}(aq),	Cr^{3+}(aq),
Mg^{2+}(aq),	Hg^{2+}(aq),	Cu^{2+}(aq),	Ag^{+}(aq),
Al^{3+}(aq),	Ni^{2+}(aq)		

Report your answer after multiplying by 10.

24. How many of the following liberate coloured vapours/gas with concentrated H_2SO_4?

(i) KBr (s) + $K_2Cr_2O_7$ (s)	(ii) KNO_2 (s)	(iii) KI (s)
(iv) K_2S (s)	(v) KCl (s)	(vi) KBr (s) + MnO_2 (s)
(vii) KNO_3 (s)	(viii) KCl(s) + MnO_2 (s)	(ix) K_2SO_3 (s)

25. Suppose that at time t the state function of a one-particle system is:

$\Psi = (2/\pi C^2)^{3/4}\, e^{-(x^2+y^2+z^2)/C^2}$ where C = 2 nm.

The probability of finding the particle at any time in the tiny cubic region with it's centre at x = 1.2 nm, y = –1.0 nm and z = 0 and with edges of each of length 0.004 nm is $x \times 10^{-y}$ (In scientific notation). Find y $\{e^{-1.22} = 0.30),\ \left(\frac{2}{4\pi}\right)^{3/2} = 0.06\}$

ANSWER KEY

1.	(ABD)	2.	(ACD)	3.	(AD)	4.	(BC)	5.	(ABCD)
6.	(ABD)	7.	(AD)	8.	(BD)	9.	(ACD)	10.	(AD)
11.	(A, B, C, D)	12.	(C, D)	13.	(A, B, D)	14.	(A)	15.	(B)
16.	(C)	17.	(16.00)	18.	(4)	19.	(6)	20.	(3)
21.	(60)	22.	(30)	23.	(70)	24.	(06)	25.	(09)